Equation Verification Log

Task 2: Equation Verification — All Chapters

Each chapter was re-read page-by-page and every equation in the corresponding knowledge file was compared against the original PDF slide. Results are recorded below.

Chapter 1 (`chap1.md` vs `chap1_HSDSD.pdf`)

Status: VERIFIED — No corrections needed.

Equation	Location in chap1.md	PDF page	Result
$f_2 = 1/(\pi\tau_r) = 0.32/\tau_r$	§1.4	p.6	(OK) Correct
Spectral breakpoints $f_1 = 1/\pi\tau$, $f_2 = 1/\pi t_r$	§1.4	p.6	(OK) Correct
$f = 0.35/t_r$, $\lambda = c/f$	§1.5	p.7	(OK) Correct
$f_l = 10f$, $\lambda_l = \lambda/10$ (TX line criterion)	§1.5	p.7	(OK) Correct

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Chapter 2 (`chap2.md` vs `chap2_HSDSD.pdf`)

Status: VERIFIED — No corrections needed.

Equation	Location	PDF page	Result
$Z_o = \sqrt{(R + j\omega L)/(G + j\omega C)}$	§1.1	p.4	(OK) Correct
Lossless $Z_o = \sqrt{L/C}$	§1.1	p.4	(OK) Correct
Microstrip $Z_o \approx \frac{87}{\sqrt{\varepsilon_r+1.41}} \ln\left(\frac{5.98H}{0.8W+T}\right)$	§2.1	p.7	(OK) Correct
Symmetric stripline $Z_{o,\text{sym}} \approx \frac{60}{\sqrt{\varepsilon_r}} \ln\left(\frac{4H}{0.67\pi(T+0.8W)}\right)$	§2.2	p.8	(OK) Correct
Propagation delay $TD = x\sqrt{\varepsilon_r}/c$	§3.1	p.10	(OK) Correct
Microstrip effective dielectric constant (full F-factor formula)	§3.2	p.11	(OK) Correct
SPICE segments $\geq 10(x/T_r v)$	§4.1	p.13	(OK) Correct
Initial wave $V_i = V_s Z_o/(Z_o + Z_s)$	§5.1	p.17	(OK) Correct

Equation	Location	PDF page	Result
Reflection coefficient $\rho = (Z_t - Z_o)/(Z_t + Z_o)$	§5.2	p.18	(OK) Correct

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Chapter 3 (`chap3.md` vs `chap3_HSDSD.pdf`)

Status: VERIFIED — No corrections needed.

Equation	Location	PDF page	Result
$V_{\text{noise},L_m} = L_m dI/dt$	§2.1	p.4	(OK) Correct
$I_{\text{noise},C_m} = C_m dV/dt$	§2.2	p.5	(OK) Correct
NEXT $A = \frac{V_{\text{in}}}{4}\left(\frac{L_M}{L} + \frac{C_M}{C}\right)$	§4.1	p.13	(OK) Correct
FEXT $B = -\frac{V_{\text{in}} X\sqrt{LC}}{2T_r}\left(\frac{L_M}{L} - \frac{C_M}{C}\right)$	§4.1	p.13	(OK) Correct
Mismatch factor $V_x = V_{\text{crosstalk}}(1 + (R - Z_0)/(R + Z_0))$	§4.2	p.15	(OK) Correct
Coupling factor $K = L_{12}/\sqrt{L_{11}L_{22}}$	§6.1	p.21	(OK) Correct
SPICE segment count $N = 10X/(vT_r)$	§6.2	p.23	(OK) Correct
Odd-mode inductance $L_{\text{odd}} = L_{11} - L_{12}$	§7.1	p.25	(OK) Correct
Odd-mode capacitance $C_{\text{odd}} = C_{11} + C_m$	§7.1	p.26	(OK) Correct
Odd-mode impedance $Z_{\text{odd}} = \sqrt{(L_{11} - L_{12})/(C_{11} + C_m)}$	§7.1	p.27	(OK) Correct
Even-mode inductance $L_{\text{even}} = L_{11} + L_m$	§7.2	p.28	(OK) Correct
Even-mode capacitance $C_{\text{even}} = C_{11} - C_m$	§7.2	p.29	(OK) Correct
Even-mode impedance $Z_{\text{even}} = \sqrt{(L_{11} + L_{12})/(C_{11} - C_m)}$	§7.2	p.29	(OK) Correct
SLEM (all in-phase) $Z_{2,\text{eff}} = \sqrt{(L_{22} + L_{12} + L_{23})/(C_{22} - C_{12} - C_{23})}$	§7.3	p.33	(OK) Correct
SLEM (one out-of-phase) $Z_{2,\text{eff}} = \sqrt{(L_{22} + L_{12} - L_{23})/(C_{22} - C_{12} + C_{23})}$	§7.4	p.34	(OK) Correct

Equation	Location	PDF page	Result
Pi-termination $R_3 = 2Z_{\text{even}}Z_{\text{odd}}/(Z_{\text{even}} - Z_{\text{odd}})$	§9.1	p.39	(OK) Correct
T-termination $R_3 = \frac{1}{2}(Z_{\text{even}} - Z_{\text{odd}})$	§9.2	p.40	(OK) Correct

Note: The example in §11.1 (p.47–49) uses a "differential mode" formula $Z_{2,\text{diff}} = \sqrt{(L_{22} - L_{12} - L_{23})/(C_{22} + C_{12} + C_{23})}$ where the center bit is out of phase with **both** neighbors simultaneously. This differs from the SLEM §7.4 (one neighbor out of phase) by design — both are correct for their respective scenarios.

Chapter 4 (`chap4.md` vs `chap4_HSDSD.pdf`)

Status: VERIFIED — No corrections needed.

Equation	Location	PDF page	Result
Even-mode TD $TD_{even} = l\sqrt{(L_s + L_m)C_s}$	§2.1/§4.2	p.10/p.23	(OK) Correct
Odd-mode TD $TD_{odd} = l\sqrt{(L_s - L_m)(C_s + 2C_m)}$	§2.1	p.10	(OK) Correct
NEXT peak $V_2(t_r) = \frac{V_s}{2}\frac{Z_{even}}{Z_{even}+Z_0} - \frac{V_s}{2}\frac{Z_{odd}}{Z_{odd}+Z_0}$	§3.5	p.14	(OK) Correct
FEXT peak $\lvert V_4(TD_{even})\rvert = V_{max}\min(T_d/t_r, 1)$	§3.5	p.14	(OK) Correct
$V_{max} = \frac{Z_0}{Z_{even}+2Z_0+Z_{odd}}V_s$	§3.5	p.14	(OK) Correct
Odd-mode TD with C_f: $TD^c_{odd} = l\sqrt{(C_s + 2C_m + 2C_f/l)(L_s - L_m)}$	§4.2	p.23	(OK) Correct
C_f design: $T_d = l\sqrt{L_s - L_m}(\sqrt{C_s + 2C_m + 2C_f/l} - \sqrt{C_s + 2C_m})$	§4.3	p.24	(OK) Correct
RC time constant $\tau = 2R_tC_f$, $R_t = Z_0Z_{odd}/(Z_0 + Z_{odd})$	§4.6	p.28	(OK) Correct
3-dB frequency $f_{3\text{dB}} = 0.35/t_r$	§5.1	p.39	(OK) Correct

Equation	Location	PDF page	Result
Guided wavelength $\lambda = (3 \times 10^8)/(f_{3\text{dB}}\sqrt{\varepsilon_{eff}})$	§5.1	p.39	(OK) Correct

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Chapter 5 (`chap5.md` vs `chap5_HSDSD.pdf`)

Status: VERIFIED — No corrections needed.

Equation	Location	PDF page	Result
DC resistance $R = \rho L/(Wt)$	§1.2	p.4	(OK) Correct
Skin depth $\delta = \sqrt{2\rho/(\omega\mu)} = \sqrt{\rho/(\pi F\mu)}$	§1.4	p.6	(OK) Correct
AC signal resistance $R_{\text{ac sig}} \approx \sqrt{\rho\pi\mu F}/W$	§1.7	p.9	(OK) Correct
DC+AC combination $R_{\text{total}} \approx \sqrt{R_{\text{ac}}^2 + R_{\text{dc}}^2}$	§1.7	p.10	(OK) Correct
Ground current density $I(D) \approx \frac{I_o}{\pi H}\frac{1}{1+(D/H)^2}$	§1.7	p.10	(OK) Correct
79.5% current integral $\int_{-3H}^{3H} \cdots = \frac{2I_o}{\pi}\tan^{-1}(3) \approx 0.795 I_o$	§1.7	p.11	(OK) Correct
Ground AC resistance $R_{\text{ac gnd}} = \sqrt{\rho\pi\mu F}/(6H)$	§1.7	p.11	(OK) Correct
Total microstrip AC resistance $\sqrt{\rho\pi\mu F}(1/W + 1/6H)$	§1.7	p.12	(OK) Correct
Exact R_{signal} (conformal mapping formula with loss ratio)	§1.7	p.12	(OK) Correct
Exact R_{ground} $((W/H)/(W/H + 5.8 + 0.03(H/W)) \cdot \sqrt{\pi\mu F\rho}/W)$	§1.7	p.12	(OK) Correct
Stripline resistance (parallel combination)	§1.8	p.13	(OK) Correct
Surface resistance $R_s = (L/W)\sqrt{\rho\pi\mu}$	§1.10	p.16	(OK) Correct

Equation	Location	PDF page	Result
Square resistance $R_{\text{square}} = \rho/t$	§1.10	p.16	(OK) Correct
Fourier series $f(x) = \frac{2}{\pi}\sum_{n=1,3,5}\frac{1}{n}\sin(2\pi nFx)$	§1.11	p.17	(OK) Correct
Attenuation $\alpha = R_2/(R_{\text{ac}} + R_2)$	§1.11	p.18	(OK) Correct
Complex dielectric $\varepsilon = \varepsilon' - j\varepsilon''$	§1.12	p.20	(OK) Correct
Loss tangent $\tan\lvert\delta_d\rvert = \varepsilon''/\varepsilon'$	§1.12	p.21	(OK) Correct
Dielectric conductance $G = (\varepsilon''/\varepsilon')(2\pi FC_{11})$	§1.12	p.21	(OK) Correct
FR4 composite $\varepsilon_r = \varepsilon_{\text{rsn}}V_{\text{rsn}} + \varepsilon_{\text{gls}}V_{\text{gls}}$	§2.2	p.29	(OK) Correct
FR4 empirical $\varepsilon_r(V_{\text{rsn}}, F) = 6.32 - [2.17 + 0.168\log_{10}F(\text{kHz})]V_{\text{rsn}}$	§2.2	p.29	(OK) Correct
90° bend capacitance $C_{90\check{r}} \approx C_{11}w$	§5.1	p.39	(OK) Correct
Excess area $= 1/4 - \pi/16 \approx 0.054$ squares	§5.1	p.39	(OK) Correct
Unbalanced T: $\rho = (Z_o/2 - Z_o)/(Z_o/2 + Z_o) = -1/3$	§6.2	p.43	(OK) Correct
Unbalanced T: $T = 1 + \rho = 2/3$	§6.2	p.43	(OK) Correct

Summary

Total equations verified: 55
Corrections made: 0
Ambiguous/flagged equations: 0

All knowledge files (`chap1.md` through `chap5.md`) accurately reflect the source PDFs. No `<!-- VERIFY: -->` flags were added because no ambiguous equations were encountered.